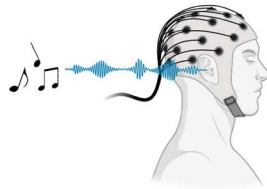


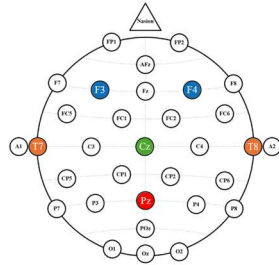
EEG-Driven Music Modulation

GitHub: /manuambrosone/EEG-Driven-Music-Modulation
Soundcloud : /manuelmbrosone/sets/eeg-drive-music-modulation

EXPERIMENT



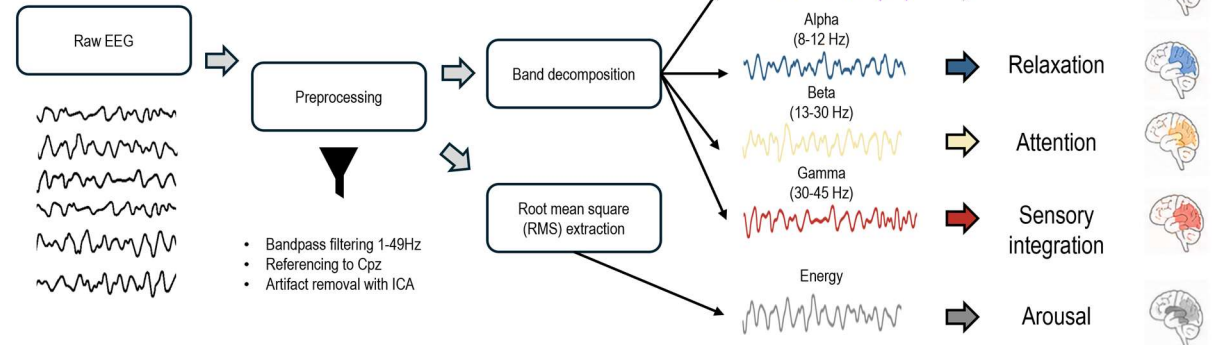
32 channels EEG during music listening



Frontal (F3, F4): attention, cognitive control
Temporal (T7, T8): auditory processing
Central (Cz): sensorimotor integration
Parietal (Pz): attention, integration

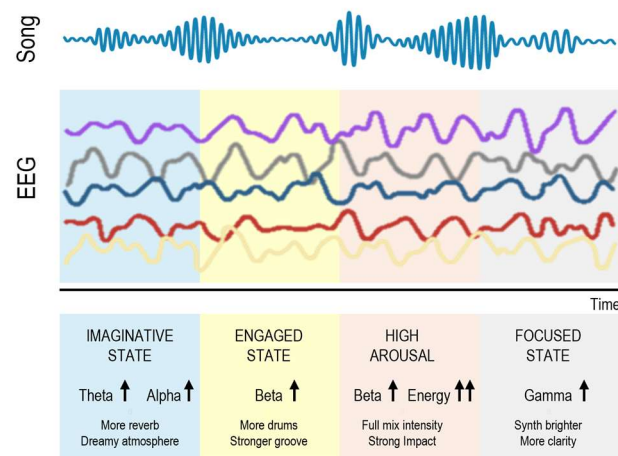
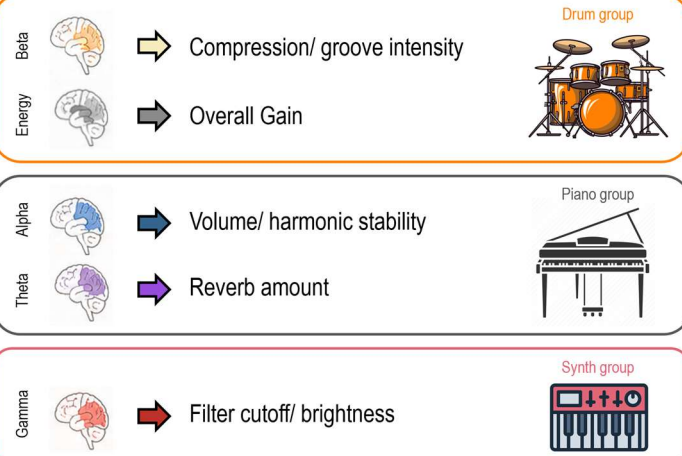
EEG captures distributed brain activity during natural music listening.

PREPROCESSING



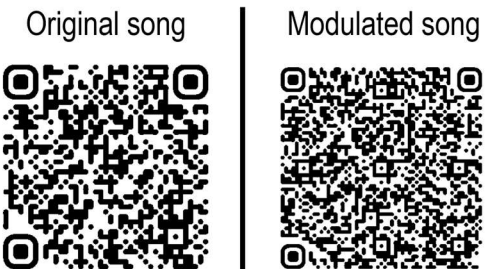
EEG activity was decomposed into neural frequency bands linked to distinct cognitive and perceptual states.

MAPPING



Different brain features drive different dimensions of the musical mix, allowing neural activity to selectively modulate rhythm, volume, spatial depth and timbral brightness.

RESULT



Scan to compare the original and EEG-modulated versions



The song is modulated by the brain, according to the listener's neural state.